Opgaver, kapitel 5

5.39 35. A block is launched with initial speed 2.5 m/s up a 30° frictionless ramp. How far up the ramp does it slide?

5.42 38. A camper hangs a 26-kg pack between two trees using separate ropes of different lengths, as shown in Fig. 5.32. Find the tension in each rope.

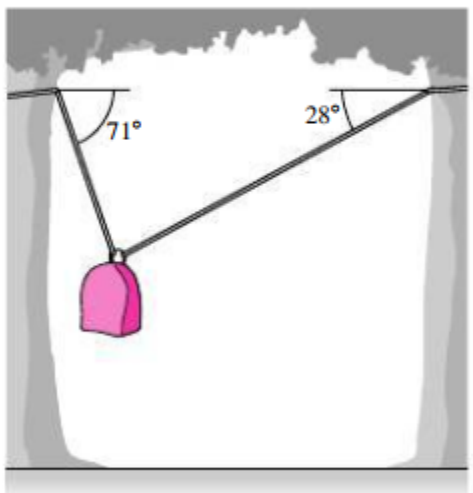

FIGURE 5.32 Problem 38

Skæve vinkler

5.47

43. When a plane turns, it banks as shown in Fig. 5.35 to give the wings' lifting force $\vec{F}_w$ a horizontal component that turns the plane. If a plane is flying level at 960 km/h and the banking angle θ is not to exceed 35°, what's the minimum curvature radius for the turn?

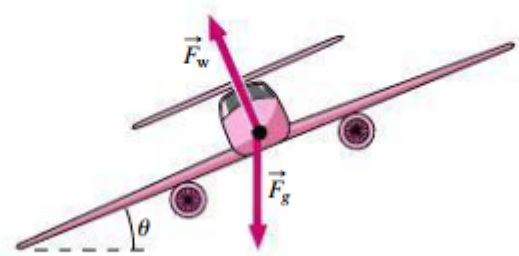

FIGURE 5.35 Problem 43

© 2016 Pearson Education, Ltd.

Opgaver, kapitel 5

5.53 49. The coefficient of static friction between steel train wheels and steel rails is 0.58. The engineer of a train moving at 140 km/h spots a stalled car on the tracks 150 m ahead. If he applies the brakes so the wheels don't slip, will the train stop in time?

Friktion

5.58 54. A police officer investigating an accident estimates that a moving car hit a stationary car at 24 km/h. Before the collision, the car left 47-m-long skid marks as it braked. The officer determines that the coefficient of kinetic friction was 0.71. What was the initial speed of the moving car?

5.71 67. A 2.1-kg mass is connected to a spring with spring constant $k = 150$ N/m and unstretched length 18 cm. The two are mounted on a frictionless air table, with the free end of the spring attached to a frictionless pivot. The mass is set into circular motion at 1.4 m/s. Find the radius of its path.

Cirkel+fjeder

© 2016 Pearson Education, Ltd.